*Islamic University of Technology*

# Lab 2: Fundamentals

CSE 4108 : Structured Programming Lab

**Section 1B**

MIA, NFS, IAC, ASH

Sep 09, 2024

# 1 Lab Tasks

## 1.1 Printing

Write a C program that uses `printf` function to print the following image.

```
III  UUU   UUU  TTTTTTTTT
III  UUU   UUU     TTT
III  UUU   UUU     TTT
III   UU   UU      TTT
III    UUUUU       TTT
```

**[Hint]** *To print backslash ('\'), you need to write double backslash ('\\') instead, since backslash is an escape character in C. Printing '\n' takes you to next line.*

## 1.2 Conversion

Two common units for measuring temperature are $Celsius(°C)$ and $Fahrenheit(°F)$. The relationship between them is as follows.

$$\frac{C}{5} = \frac{F - 32}{9}$$

Write a C program that would convert 37$°C$ to Fahrenheit.

## 1.3 Dynamic Input

Change the program from 1.2 to take the input value from user in Celsius and convert to Fahrenheit.

**[Sample Program]**

```
Enter value in Celsius: 37
Value in Fahrenheit: 98.6
```

## 1.4 Hydrocarbon: Alcohol

In chemistry, alcohol is an organic compound that carries at least one hydroxyl functional group (-OH) bound to a saturated carbon atom. An important class of alcohols, of which methanol and ethanol are the simplest members, includes all compounds for which the general formula is $C_nH_{2n+1}OH$.

Write a C program that would take the value of $n$ as input and print the corresponding simple alcohol formula.

**[Sample Program]**

```
Enter value of n: 2
Formula: C2H5OH
```

### 1.5 Equation Evaluation

Write a C program to take the value of $x$ as input and evaluate the value of $y$ in the following polynomial equation.

$$y = \frac{3x^3 + 4x^2 - 2.5x}{(1.3x^3 + 5) * 3}$$

**[Sample Program]**

```
Enter value of x: 3.5
Value of y: 0.926803
```

**[Hint]** *C has no exponential operator directly in the language. So it needs to be done using multiplications. For instance,* $x^3 = x * x * x$. *Negative exponent can be represented using division and positive exponent. For instance,* $x^{-3} = \frac{1}{x^3}$.

### 1.6 Counting Geese

Mr. Poached Eggsy is the owner of a magical goose farm. A magical goose only lays **13 platinum eggs** at the beginning of the year. The eggs take 6 months to hatch and another 6 months are required for the goslings to mature. So, the new geese **can lay eggs right after one year**. However, Mr. Eggsy sells **half of all the eggs** each year to make his ends meet. In case he has odd number of eggs, e.g. **15, he will sell 7 eggs and keep 8 eggs** to increase the number of geese in his farm.

Given the number of geese $n$ at the beginning of first year as input, calculate how many geese Mr. Eggsy will have after the $1^{st}$ year, $2^{nd}$ year and $3^{rd}$ year.

**[Sample Program]**

```
Enter initial geese count: 3
Geese count after year 1: 23
Geese count after year 2: 173
Geese count after year 3: 1298
```

**[Hint]** *Be sure to use integers for all the values. Dividing integer by integer will always result in integer. For instance,* $7/2 = 3$. *In these odd cases, the other half can be calculated by subtracting first half from total, i.e* $7 - 3 = 4$. *If a problem seems difficult, first try to break it down and do it on pen and paper. Once the actual logic is clear to you, then begin coding This process, in some cases, is referred to as 'Dry Run'.*

## 2 Sample C Program

```c
/* Author: Mohammad Ishrak Abedin
 * Topic Review | Examples of things done so far
 */

#include <stdio.h> // This is a "Preprocessor Directive" to include standard I/O libary for scanning and printing

#define PI 3.14159265 // This is a macro defined constant

```

```c
int main(void) // Every C program must have a "main"(case sensitive) function as entry point
{
    int myInt1, myInt2; // Declaration of two variables of type Integer
    myInt1 = 2021; // Assignment of variable myInt1
    myInt2 = -2021; // Assignment of variable myInt2, int accepts negative values
    float myFloat = 20.5f; // Declartion and assignment of a variable of type Floating Point

    int inputVariable; // Another variable that will be used to hold an input

    printf("Please input an integer number: "); // A print statement that prints to the "console"
    scanf("%d", &inputVariable); // Takes input and stores it in inputVariable as integers

    // The following statement prints all the variables used so far
    printf("myInt1: %d\nmyInt2: %d\nmyFloat: %f\ninputVariable: %d\n", myInt1, myInt2, myFloat, inputVariable);

    printf("PI: %f", PI);

    return 0; // A return statement, that exits main function and returns the control back (where? will be explained later)
}

// C99
```

Listing 1: Sample C Program